

List of Publications — Prof. Dr. Boris Vertman

5 most important contributions

- A. *Signatures of Witt spaces with boundary*
with Piazza, **Advances Math.** **405** (2022)
- B. *Eta and rho invariants on manifolds with edges*
with Paolo Piazza, **Annales Inst. Fourier** **69**, no. 5 (2019)
- C. *Stability of Ricci de Turck flow on Singular Spaces*
with Klaus Kröncke, **Calc. Var. PDE** **58**, no. 2 (2019)
- D. *Regularizing infinite sums of zeta-determinants*
with Matthias Lesch, **Math. Annalen** **361** no. 3-4 (2015)
- E. *Analytic torsion on manifolds with edges*
with Rafe Mazzeo, **Advances in Math.** **231** (2012)

Preprints in preparation

- (53). *Microlocal analysis and discrete geometry*
with Nicolai Reshetikhin (2025)
- (52). *Positive mass theorem on hyperbolic spaces with boundary*
with Matheus Santos (2025)

Submitted preprints

- (51). *Generalized Yamabe flows*
with Lye and Vitti, arXiv:2412.02593 [math.DG]
- (50). *L^2 Gamma index theorem for space times*
with Damaschke, arXiv:2410.05848 [math.DG]
- (49). *Scattering theory for difference equations with operator coefficients*
with Silva, Sher and Winklmeier, arXiv:2501.11194 [math.SP]

Publications (accepted or published)

I. Spectral invariants on singular spaces

48. *Analytic torsion on manifolds with fibered boundary metrics*
with Lye, **IMRN**, Vol. **2025**, no. 1 (2025)
47. *Cheeger-Müller theorem for a wedge singularity along an embedded submanifold*, with Hartmann, **Rev. Mat. Complut.** (2024)
46. *Stability of L^2 -invariants on stratified spaces*
with Bei and Piazza, **IMRN**, Vol. **2024**, no. 21 (2024)
45. *Signatures of Witt spaces with boundary*
with Piazza, **Advances Math.** **405** (2022)

44. *Eta and rho invariants on manifolds with edges* with Piazza, **Ann. Inst. Fourier** **69**, no. 5 (2019)
43. *Cheeger-Müller theorem on manifolds with cusps* **Math. Zeitschrift** **291**, no. 3-4 (2019)
42. *Zeta-determinants of Sturm-Liouville operators with quadratic potentials at infinity*, with Hartmann and Lesch, **Journal of Diff. Equations** **262**, no. 5 (2017)
41. *Refined analytic torsion as analytic function on the representation variety and applications*, with Braverman, **Annales Math. du Quebec** **41**, no. 1 (2017)
40. *Regularizing infinite sums of zeta-determinants* with Lesch, **Math. Annalen** **361** 3-4, (2015)
39. *The Metric Anomaly of Analytic Torsion on Manifolds with Conical Singularities*, with Müller, **Communications in PDE**. **39**, no. 1 (2014)
38. *A new proof of a Bismut-Zhang formula for some class of representations*, with Braverman, **Proc. Contemp. Math.** **630** (2014)
37. *Analytic torsion on manifolds with edges* with Mazzeo, **Advances in Math.** **231** (2012)
36. *The metric anomaly of analytic torsion at the boundary of an even dimensional cone*, **Ann. Geom. Anal. Geom.** **41** no. 1 (2011)
35. *Regular singular Sturm-Liouville operators and their zeta-determinants*, with Lesch, **J. Funct. Anal.** **261** (2) (2011)
34. *Recent Advances on the Analytic Torsion of Singular Spaces* Oberwolfach Report, "Analysis and Geometric Singularities" (2010)
33. *Refined analytic torsion on manifolds with boundary* **Geometry and Topology** **13**, no. 4 (2009)
32. *Analytic Torsion of a Bounded Generalized Cone* **Comm. in Mathematical Physics** **290**, no. 3 (2009)
31. *Zeta determinants for regular-singular Laplace-type operators* **Journal of Mathematical Physics** **50**, no. 8 (2009)
30. *Exotic Expansions and pathological properties of zeta-functions on conic manifolds*, Kirsten, Loya, Park with an Appendix by Vertman, **Journal of Geometric Analysis** **18** (2008)

II. Geometric flows, non-linear PDE on singular spaces

29. *Phase transitions and minimal interfaces on manifolds with conical singularities*, with Grieser, Held and Uecker, **Nonlinearity** **38**, no. 2 (2025)
28. *Normalized Yamabe flow on manifolds with bounded geometry* with Caldeira and Hartmann, **Ann. Global Anal. Geom.** **63**, no. 3 (2023)

27. *Convergence of the Yamabe flow on singular spaces with positive Yamabe constant*, with Carron and Lye, **Tohoku J. Math. (2)** **75**, no. 4 (2023)
26. *Long-time existence of Yamabe flow on singular spaces with positive Yamabe constant*, with Lye, **Anal. PDE** **16**, no. 2 (2023)
25. *Bounded Ricci Curvature and Positive Scalar Curvature under Singular Ricci de Turck Flow*, with Kröncke and Marxen **Pacific J. Math.** **324**, nr. 2, (2023)
24. *Ricci de Turck flow on incomplete manifolds with Marxen*, **Documenta Math.** **27**, (2022)
23. *A Survey on the Ricci flow on Singular Spaces with Kröncke*, *Diff. geometry in the large*, 118–139 **London Math. Soc. Lecture Note Ser.**, **463**, (2021)
22. *Perelman's entropies for manifolds with conical singularities with Kröncke*, **Trans. Amer. Math. Soc.** **374** no. 4 (2021)
21. *Ricci flow on singular manifolds* **J. Geom. Anal.** **31**, no. 4 (2021)
20. *Stability of Ricci de Turck flow on Singular Spaces with Kröncke*, **Calc. Var. PDE** **58**, no. 2 (2019)
19. *Long-time existence of the edge Yamabe flow with Bahuaud*, **J. Math. Soc. Japan** **71**, no. 2 (2019)
18. *Yamabe flow on manifolds with edges with Bahuaud*, **Math. Nachrichten** **287**, no. 23 (2014)

III. Mathematical Physics

17. *Prescribed mean curvature flow of non-compact space-like Cauchy hypersurfaces*, with Gentile, **Ann. Global Anal. Geom.** **64**, no. 2 (2023)
16. *Mean curvature flow in asymptotically flat product spacetimes with Kröncke, Petersen, Lubbe, Marxen, Maurer, Meiser, Schnürer, Szabo* **J. Geom. Analysis** **31** (2021)
15. *Combinatorial Quantum Field Theory and Gluing Formula for Determinants*, with Reshetikhin **Letters in Math. Physics** **105**, no. 3 (2015)

IV. Discrete geometry

14. *Extremals of determinants of Laplacians on discrete surfaces with Hafemann*, **Adv. Th. Math. Phys.** **28**, no. 5 (2024)
13. *Spectral zeta function on discrete tori and Epstein-Riemann conjecture with Meiners*, **J. Number Theory** **244**, (2023)
12. *Regularized limit of determinants for discrete tori* **Monatshefte für Mathematik** **186**, no. 3 (2018)

V. Singular PDE, Ψ DO and boundary value problems

11. *Spectral geometry on manifolds with fibred boundary metrics II: heat kernel asymptotics*, with Talebi, **Anal. Math. Phys.** **12**, no. 2 (2022)
10. *Spectral geometry on manifolds with fibred boundary metrics I: Low energy resolvent*, with Talebi and Grieser
J. l'École Polytechnique - Math. **9**, (2022)
9. *Resolvent trace asymptotics on stratified spaces*
with Hartmann and Lesch, **Pure Appl. Anal.** **3** no. 1 (2021)
8. *On the domain of Dirac and Laplace type Operators on Stratified Spaces*
with Hartmann and Lesch, **Journal of Spectral Theory**, **8**, no. 4 (2018)
7. *The biharmonic heat operator on edge manifolds and non-linear fourth order equations*, **Manuscripta Mathematica** **149** no. 1 (2016)
6. *Multiparameter resolvent trace expansion for elliptic boundary problems*
Analysis, Geometry and Number theory J. **1** (2016)
5. *Mapping properties of the heat operator on edge manifolds*
with Dryden and Bahuaud **Math. Nachr.** **288**, no. 2 (2015)
4. *Heat trace asymptotics for edge Laplacians with algebraic boundary conditions*, **Journal d'Analyse Math.** **125**, no. 1 (2015)
3. *Elliptic theory of differential edge operators, II: Boundary value problems*, with Mazzeo, **Indiana Math. Journal** **63**, no. 6 (2014)
2. *The exotic heat-trace asymptotics of a regular-singular operator revisited*
Journal of Math. Physics **54**, no. 6 (2013)

VI. Doctoral thesis

1. *Analytic Torsion of Manifolds with Boundary and Conical Singularities*, **Bonn University Press** (2008)